Multiple IV Infusions & Multiple Oral Practice 3

Module 6 - Multiple Dosings · Daily practice sheet · No date

Name _____ Date _____

1a 1. A 200-mg dose of an antibiotic was administered as an IV infusion over a period of 2 hours. Eight hours after the start of the first infusion, a second dose of 200 mg was infused, again over 2 hours. The antibiotic has a half-life of 4 hours and a total body clearance of 2.8 L/hr. a. What is the plasma drug concentration at the end of the first infusion?

1b 1. A 200-mg dose of an antibiotic was administered as an IV infusion over a period of 2 hours. Eight hours after the start of the first infusion, a second dose of 200 mg was infused, again over 2 hours. The antibiotic has a half-life of 4 hours and a total body clearance of 2.8 L/hr. b. What is the plasma drug concentration 4 hours after the cessation of the second infusion?

2a 2. A patient is receiving his antihypertensive medication as 500 mg every 8 hours in the form of an oral capsule. The bioavailability of the drug from this capsule is 80% and the total body clearance in this patient is 2.5 L/hr. a. What is the average steady-state plasma concentration of this drug in this patient?

2b 2. A patient is receiving his antihypertensive medication as 500 mg every 8 hours in the form of an oral capsule. The bioavailability of the drug from this capsule is 80% and the total body clearance in this patient is 2.5 L/hr. b. What will be the average steady-state plasma concentration if the patient were taking 1000 mg every 8 hours of the same oral capsule?

2c 2. A patient is receiving his antihypertensive medication as 500 mg every 8 hours in the form of an oral capsule. The bioavailability of the drug from this capsule is 80% and the total body clearance in this patient is 2.5 L/hr. c. What is the Cl_T of the drug in this patient while taking the 1000 mg orally every 8 hours?

A. 2.0 L/hr, the clearance times F
B. 2.5 L/hr, unchanged
C. 1.25 L/hr, half the clearance
D. 5 L/hr, twice the clearance

2d 2. A patient is receiving his antihypertensive medication as 500 mg every 8 hours in the form of an oral capsule. The bioavailability of the drug from this capsule is 80% and the total body clearance in this patient is 2.5 L/hr. d. What will be the average steady-state plasma concentration if the patient were taking 500 mg every 6 hours of the same oral capsule?

2e 2. A patient is receiving his antihypertensive medication as 500 mg every 8 hours in the form of an oral capsule. The bioavailability of the drug from this capsule is 80% and the total body clearance in this patient is 2.5 L/hr. e. What oral dose is required to reach an average steady-state plasma concentration of 30 mg/L from this drug product in this patient taken every 8 hours?

2f 2. A patient is receiving his antihypertensive medication as 500 mg every 8 hours in the form of an oral capsule. The bioavailability of the drug from this capsule is 80% and the total body clearance in this patient is 2.5 L/hr. f. Because the patient had problems swallowing the oral capsules he was shifted to 500 mg IV every 8 hours. What will be the average steady-state concentration of this drug in this patient?